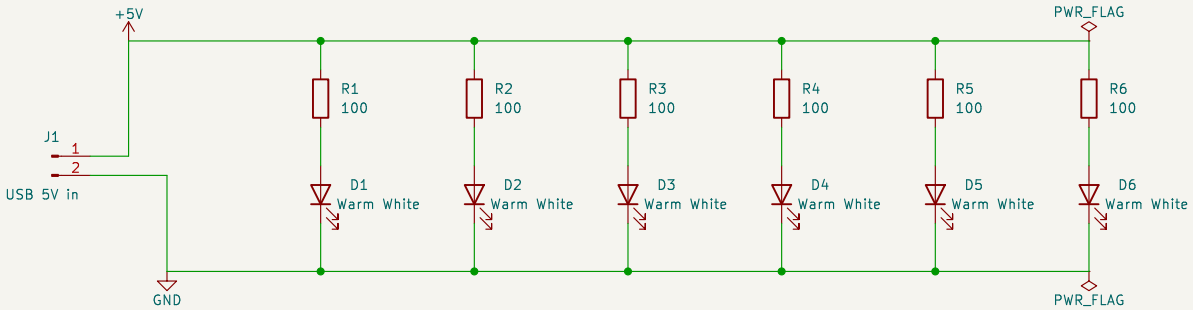
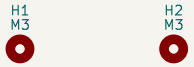
USB cable wires:
  red   -> +5V (J1 pin 1)
  black -> GND (J1 pin 2)
(data wires unused — cut them back)

Each LED has its own resistor (6 parallel branches).
I(LED) ≈ (5.0 V − 2.9 V) / 100 Ω ≈ 20 mA (typical LED-strip drive current)
Total ≈ 125 mA from USB. P(R) ≈ 44 mW (0805 rating 125 mW).
D1–D6: XINGLIGHT XL-2835WWC-02 warm white 2835 (2800–3000 K), LCSC C2843875
R1–R6: UNI-ROYAL 0805W8F1000T5E 100R 1% 0805, LCSC C17408



Mounting holes (M3, no connection)



Assembled by JLCPCB (economic PCBA, top side)
6x warm white 2835 LED light bar, 100 x 15 mm, powered from a cut 5 V USB cable

Sheet: /
File: usb-led-bar.kicad_sch

Title: USB LED Bar

Size: A4	Date: 2026-10-15	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1